

1. Peran data warehouse dan data lake dalam pengambilan keputusan

Aspek	Data Warehouse	Data Lake
Tujuan	Mendukung analisis dan pelaporan bisnis terstruktur	Menyimpan data mentah untuk eksplorasi, ML, dan big data
Jenis Data	Terstruktur (tabel, kolom, angka)	Terstruktur, semi terstruktur (JSON, CSV), dan tidak terstruktur (gambar, teks)
Pengguna Utama	Analisis bisnis, manajemen	Data scientist, engineer
Proses Data	Sudah dibersihkan, diubah, dan diintegrasikan melalui ETL	Data disimpan dalam bentuk mentah (ELT atau schema on read)
Kegunaan	BI, laporan, dashboard	Eksperimen, analisis prediktif, dan ML
Waktu pemilihannya	Data yang digunakan untuk <i>reporting</i> rutin, BI, dan analisis historis. Dibutuhkan <i>data konsisten</i> dengan kualitas tinggi. Perusahaan fokus pada performa kueri dan laporan periodik.	Organisasi ingin menyimpan <i>semua jenis data</i> (mentah, teks, log, sensor, dsb). Ada rencana penggunaan <i>machine learning</i> atau analisis prediktif. Perusahaan belum tahu semua kebutuhan analisisnya di awal (<i>schema-on-read</i>).

2. Desain star schema untuk DW rumah sakit (ini general)

Fakta yg bisa dianalisis:

Jumlah pasien, biaya perawatan, penggunaan obat, hubungan gejala dan perawatan

Tabel Fakta: Fact_treatment

Kolom	Keterangan
treatment_id	Kunci utama fakta
patient_id	FK ke dim pasien
doctor_id	FK ke dim dokter
department_id	FK ke dim departemen
diagnosis_id	FK ke dim diagnosis
medication_id	FK ke dim obat
treatment_date	FK ke dim waktu
total_cost	Jumlah biaya perawatan
length_of_stay	Lama rawat (jika ada)

Tabel Dimensi:

1. **Dim_Patient**(patient_id, gender, age, insurance_type)
2. **Dim_Doctor**(doctor_id, name, specialization)
3. **Dim_Department**(department_id, name, location)
4. **Dim_Diagnosis**(diagnosis_id, diagnosis_name, category)
5. **Dim_Medication**(medication_id, medication_name, drug_type)
6. **Dim_Treatment**(treatment_id, treatment_type, description)
7. **Dim_Time**(time_id, date, month, quarter, year)

Star schema:

3. Desain star schema untuk DW rumah sakit (ini spesifik)

a) Analisis performa dept: jumlah pasien & total biaya perawatan per kuartal

Fact table: Fact_department_performance

Kolom	Keterangan
fact_id	PK
department_id	FK ke dim_department
time_id	FK ke dim_time
total_patients	Jumlah pasien
total_cost	Total biaya perawatan

Tabel dimensi:

1. Dim_dept: dept_id, dept_name, location
2. Dim_time: time_id, date, month, quarter, year

Star scheme

b) Analisis pengguna obat berdasarkan penyakit

Fact table: fact_med_usage

Kolom	Keterangan
fact_id	PK
diagnosis_id	FK ke dim_diagnosis
medication_id	FK ke dim_medication
time_id	FK ke dim_time
total_medication_used	Jumlah obat yang digunakan
total_medication_cost	Biaya obat total

Dimensi:

1. Dim_Diagnosis (diagnosis_id, diagnosis_name, category)
2. Dim_Medication (medication_id, medication_name, drug_type)
3. Dim_Time (time_id, date, month, quarter, year)

Star scheme:

c) Analisis hubungan gejala dan jenis perawatan

Fact table: fact_symptom_treatment

Kolom	Keterangan
fact_id	PK
symptom_id	FK ke dim_symptom
treatment_id	FK ke dim_treatment
time_id	FK ke dim_time
total_patients	Jumlah pasien dengan kombinasi tsb
avg_cost	Rata-rata biaya perawatan

Dimensi:

1. Dim_Symptom (symptom_id, symptom_name, category)
2. Dim_Treatment (treatment_id, treatment_type, description)
3. Dim_Time (time_id, date, month, quarter, year)

Star schema:

4. Contoh query multidimensi (OLAP SQL)

a) Analisis performa departemen

```
SELECT
    d.department_name,
    t.year,
    t.quarter,
    COUNT(f.patient_id) AS total_patients,
    SUM(f.total_cost) AS total_cost
FROM Fact_Treatment f
JOIN Dim_Department d ON f.department_id = d.department_id
JOIN Dim_Time t ON f.treatment_date = t.time_id
GROUP BY d.department_name, t.year, t.quarter
ORDER BY t.year, t.quarter;
```

b) Analisis pengguna obat berdasarkan diagnosis

```
SELECT
    diag.diagnosis_name,
    med.medication_name,
    SUM(f.total_cost) AS total_medication_cost
FROM Fact_Treatment f
JOIN Dim_Medication med ON f.medication_id = med.medication_id
JOIN Dim_Diagnosis diag ON f.diagnosis_id = diag.diagnosis_id
GROUP BY diag.diagnosis_name, med.medication_name;
```

c) Hubungan gejala dan jenis perawatan

```
SELECT
    diag.diagnosis_name,
    treat.treatment_type,
    AVG(f.total_cost) AS avg_treatment_cost
FROM Fact_Treatment f
JOIN Dim_Diagnosis diag ON f.diagnosis_id = diag.diagnosis_id
JOIN Dim_Treatment treat ON f.treatment_id = treat.treatment_id
GROUP BY diag.diagnosis_name, treat.treatment_type;
```

5. Contoh proses ETL (ini yang umumnya)

a) Extract (ambil data dari database operasional kaya tabel admissions, med_records, transaction)

```
SELECT
    a.admission_id, a.patient_id, a.doctor_id, a.admission_date,
    m.diagnosis, t.amount AS total_cost
FROM Admissions a
JOIN Medical_Records m ON a.admission_id = m.admission_id
JOIN Transactions t ON a.admission_id = t.admission_id;
```

b) Transform (membersihkan dan mengubah data kaya itung umur pasien, membuat kolom waktu)

```
SELECT
    a.patient_id,
    DATEDIFF(YEAR, p.date_of_birth, GETDATE()) AS age,
    YEAR(a.admission_date) AS year,
    DATEPART(QUARTER, a.admission_date) AS quarter,
    t.amount AS total_cost
FROM Admissions a
JOIN Patients p ON a.patient_id = p.patient_id
JOIN Transactions t ON a.admission_id = t.admission_id;
```

c) Load (memasukkan hasil transformasi ke tabel fakta di DW)

```
INSERT INTO Fact_Treatment (patient_id, doctor_id, department_id,
diagnosis_id, medication_id, treatment_date, total_cost)
SELECT
    a.patient_id, a.doctor_id, d.department_id, m.record_id, med.medication_id,
    a.admission_date, t.amount
FROM Admissions a
JOIN Doctors d ON a.doctor_id = d.doctor_id
JOIN Medical_Records m ON a.admission_id = m.admission_id
JOIN Medications med ON m.record_id = med.record_id
JOIN Transactions t ON a.admission_id = t.admission_id;
```

6. Contoh proses ETL (ini yang spesifiknya)

a) ETL analisis performance

```
SELECT
    a.admission_id,
    a.department_id,
    a.admission_date,
    t.amount AS total_cost,
    a.patient_id
FROM Admissions a
JOIN Transactions t ON a.admission_id = t.admission_id;
```

```
SELECT
    a.department_id,
    YEAR(a.admission_date) AS year,
    DATEPART(QUARTER, a.admission_date) AS quarter,
    COUNT(DISTINCT a.patient_id) AS total_patients,
    SUM(t.amount) AS total_cost
FROM Admissions a
JOIN Transactions t ON a.admission_id = t.admission_id
GROUP BY a.department_id, YEAR(a.admission_date), DATEPART(QUARTER,
a.admission_date);
```

```
INSERT INTO Fact_Department_Performance (department_id, time_id,
total_patients, total_cost)
```

```
SELECT
    d.department_id,
    t.time_id,
    agg.total_patients,
    agg.total_cost
FROM (
    SELECT
        a.department_id,
        YEAR(a.admission_date) AS year,
        DATEPART(QUARTER, a.admission_date) AS quarter,
        COUNT(DISTINCT a.patient_id) AS total_patients,
        SUM(t.amount) AS total_cost
    FROM Admissions a
    JOIN Transactions t ON a.admission_id = t.admission_id
    GROUP BY a.department_id, YEAR(a.admission_date),
DATEPART(QUARTER, a.admission_date)
) agg
JOIN Dim_Department d ON d.department_id = agg.department_id
JOIN Dim_Time t ON t.year = agg.year AND t.quarter = agg.quarter;
```

b) ETL analisis pengguna obat

```
SELECT
    m.record_id,
    m.diagnosis,
    md.medication_id,
    md.dosage,
    t.amount AS medication_cost,
    m.doctor_id,
    m.admission_id
FROM Medical_Records m
JOIN Medications md ON m.record_id = md.record_id
JOIN Transactions t ON m.admission_id = t.admission_id;
```

```
SELECT
    m.diagnosis AS diagnosis_id,
    md.medication_id,
    YEAR(m.admission_date) AS year,
    DATEPART(QUARTER, m.admission_date) AS quarter,
    COUNT(md.medication_id) AS total_medication_used,
    SUM(t.amount) AS total_medication_cost
FROM Medical_Records m
JOIN Medications md ON m.record_id = md.record_id
JOIN Admissions a ON m.admission_id = a.admission_id
JOIN Transactions t ON a.admission_id = t.admission_id
GROUP BY m.diagnosis, md.medication_id, YEAR(m.admission_date),
DATEPART(QUARTER, m.admission_date);
```

```
INSERT INTO Fact_Medication_Usage (diagnosis_id, medication_id, time_id,
total_medication_used, total_medication_cost)
```

```
SELECT
    dgn.diagnosis_id,
    med.medication_id,
    t.time_id,
    agg.total_medication_used,
    agg.total_medication_cost
FROM (
    SELECT
        m.diagnosis AS diagnosis_id,
        md.medication_id,
        YEAR(m.admission_date) AS year,
        DATEPART(QUARTER, m.admission_date) AS quarter,
        COUNT(md.medication_id) AS total_medication_used,
        SUM(t.amount) AS total_medication_cost
    FROM Medical_Records m
    JOIN Medications md ON m.record_id = md.record_id
    JOIN Admissions a ON m.admission_id = a.admission_id
    JOIN Transactions t ON a.admission_id = t.admission_id
    GROUP BY m.diagnosis, md.medication_id, YEAR(m.admission_date),
DATEPART(QUARTER, m.admission_date)
) agg
```



```

JOIN Dim_Diagnosis dgn ON dgn.diagnosis_id = agg.diagnosis_id
JOIN Dim_Medication med ON med.medication_id = agg.medication_id
JOIN Dim_Time t ON t.year = agg.year AND t.quarter = agg.quarter;

```

c) ETL analisis hubungan gejala

```

SELECT
    s.symptom_id,
    tmt.treatment_id,
    a.admission_date,
    tr.amount AS total_cost,
    a.patient_id
FROM Symptoms s
JOIN Treatments tmt ON s.record_id = tmt.record_id
JOIN Admissions a ON s.record_id = a.admission_id
JOIN Transactions tr ON a.admission_id = tr.admission_id;

SELECT
    s.symptom_id,
    tmt.treatment_id,
    YEAR(a.admission_date) AS year,
    DATEPART(QUARTER, a.admission_date) AS quarter,
    COUNT(DISTINCT a.patient_id) AS total_patients,
    AVG(tr.amount) AS avg_cost
FROM Symptoms s
JOIN Treatments tmt ON s.record_id = tmt.record_id
JOIN Admissions a ON s.record_id = a.admission_id
JOIN Transactions tr ON a.admission_id = tr.admission_id
GROUP BY s.symptom_id, tmt.treatment_id, YEAR(a.admission_date),
DATEPART(QUARTER, a.admission_date);

INSERT INTO Fact_Symptom_Treatment (symptom_id, treatment_id, time_id,
total_patients, avg_cost)
SELECT
    sym.symptom_id,
    trt.treatment_id,
    t.time_id,
    agg.total_patients,
    agg.avg_cost
FROM (
    SELECT
        s.symptom_id,
        tmt.treatment_id,
        YEAR(a.admission_date) AS year,
        DATEPART(QUARTER, a.admission_date) AS quarter,
        COUNT(DISTINCT a.patient_id) AS total_patients,
        AVG(tr.amount) AS avg_cost
    FROM Symptoms s
    JOIN Treatments tmt ON s.record_id = tmt.record_id
    JOIN Admissions a ON s.record_id = a.admission_id
    JOIN Transactions tr ON a.admission_id = tr.admission_id

```

```
GROUP BY s.symptom_id, tmt.treatment_id, YEAR(a.admission_date),  
DATEPART(QUARTER, a.admission_date)  
) agg  
JOIN Dim_Symptom sym ON sym.symptom_id = agg.symptom_id  
JOIN Dim_Treatment trt ON trt.treatment_id = agg.treatment_id  
JOIN Dim_Time t ON t.year = agg.year AND t.quarter = agg.quarter;
```

2. Lakukan proses yang diperlukan untuk mendapatkan informasi:

1. Daftar negara yang jumlah keuntungannya lebih besar dari rata jumlah keuntungan regionnya?
2. Produk yang paling laku di masing-masing negara produk apa?
3. Rerata waktu pemrosesan (selisih antara pemesanan hingga pengiriman) per tipe item di setiap region
4. Region manakah yang berhasil memperoleh angka jumlah keuntungan penjualan tertinggi pada masing-masing tahun?

1. Negara yg untung > region

```
SELECT Negara, Wilayah, Keuntungan
FROM (
  SELECT Negara,
    Wilayah,
    SUM(UnitTerjual*(HargaSatuan-BiayaProduksiSatuan)) AS Keuntungan,
    AVG(SUM(UnitTerjual*(HargaSatuan-BiayaProduksiSatuan))) OVER (PARTITION BY
Wilayah) AS AvgKeuntunganWilayah
  FROM LatihanDW
  GROUP BY Negara, Wilayah
) t
WHERE Keuntungan > AvgKeuntunganWilayah;
```

2. Produk paling laku

```
SELECT Negara, JenisBarang, TotalUnitTerjual
FROM (
  SELECT Negara, JenisBarang,
    SUM(UnitTerjual) AS TotalUnitTerjual,
    RANK() OVER (PARTITION BY Negara ORDER BY SUM(UnitTerjual) DESC) AS rnk
  FROM LatihanDW
  GROUP BY Negara, JenisBarang
) x
WHERE rnk = 1;
```

3. Region yg memperoleh angka keuntungan penjualan tertinggi masing2 tahun!

```
SELECT Tahun,
  Wilayah,
  TotalKeuntungan
FROM (
  SELECT EXTRACT(YEAR FROM t."TglPesan") AS Tahun,
    t.Wilayah,
    SUM(t."UnitTerjual" * (t."HargaSatuan" - t."BiayaProduksiSatuan")) AS TotalKeuntungan,
    RANK() OVER (PARTITION BY EXTRACT(YEAR FROM t."TglPesan")
      ORDER BY SUM(t."UnitTerjual" * (t."HargaSatuan" - t."BiayaProduksiSatuan"))
DESC) AS urutan
  FROM TugasDW t
  GROUP BY Tahun, t.Wilayah
) ranked
WHERE urutan = 1
ORDER BY Tahun ASC;
```

4. Negara keuntungan tertinggi, dan keuntungannya diperoleh tahun berapa?

```
SELECT Negara,  
       Tahun,  
       TotalKeuntungan  
FROM (  
    SELECT t.Negara,  
           EXTRACT(YEAR FROM t."TglPesan") AS Tahun,  
           SUM(t."UnitTerjual" * (t."HargaSatuan" - t."BiayaProduksiSatuan")) AS TotalKeuntungan,  
           RANK() OVER (ORDER BY SUM(t."UnitTerjual" * (t."HargaSatuan" -  
t."BiayaProduksiSatuan")) DESC) AS urutan  
    FROM TugasDW as t  
    GROUP BY t.Negara, Tahun  
    ) ranked  
WHERE urutan = 1;
```